Lecture 6

Signaling

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Asymmetric Information and Extensive Forms

In many interesting applications, one party is better informed than another

A particular form of this setting is the basic signaling game:

- Two actors: sender and receiver
- Informed sender gets to take an action which is observed by uninformed receiver
- Receiver attempts to deduce state from sender's action, makes decision that affects both players

Can be modeled as an extensive form game in which sender has several possible types (one for each value of the state)

Entry and Signaling

Country A is attempting to take land from country B

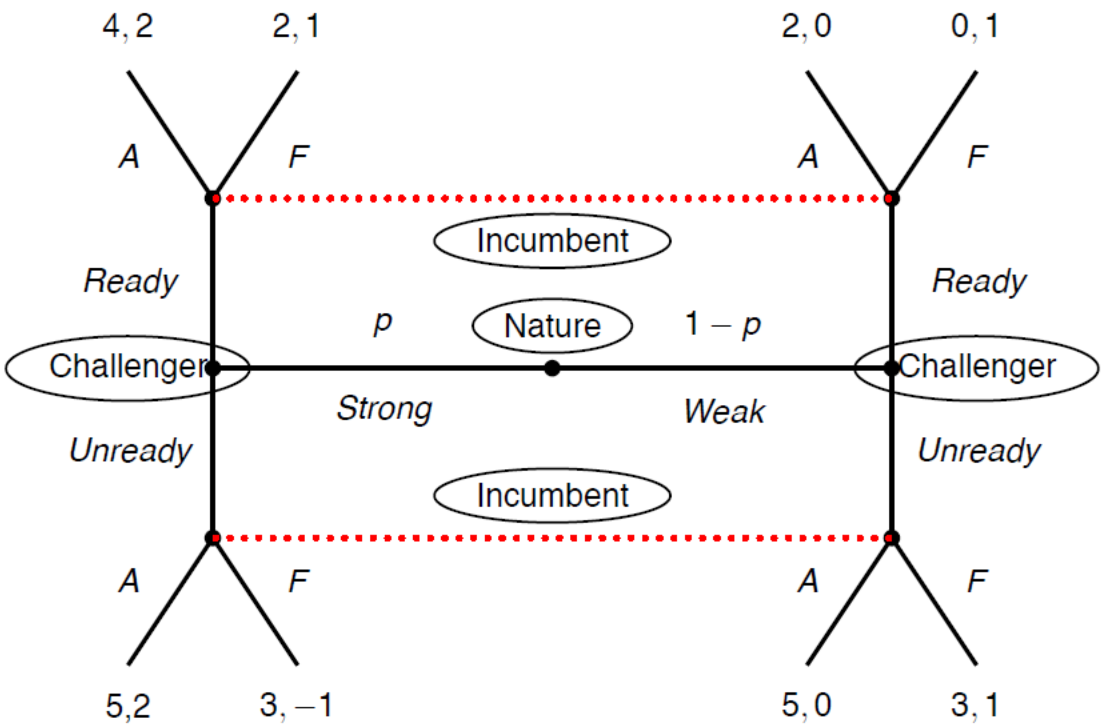
A is **strong** with probability p and **weak** with probability $1 - p$

A knows its own type, but Country B does not

A may either *ready* itself for war or remain *unready*

B observes A's readiness, but not its type, and chooses whether to *fight* or *acquiesce*

The Crisis Bargaining Game



PBE in the Crisis Bargaining Game

In any PBE, Weak Challenger chooses *Unready*

Suppose Strong Challenger Plays *Ready*

- Incumbent learns Challengers type by observing action
- Let $\mu(A)$ be probability Incumbent assigns to strong, following action A: $\mu(R) = 1$ and $\mu(U) = 0$
- Incumbent chooses A if R and chooses F if U
- Strong type, if deviates to U, gets 3 rather than 4.
- Weak type, if deviates to R gets 2 rather than 3
- A PBE

Suppose Strong Challenger chooses *Unready*

Incumbent learns nothing from observing U: $\mu(U) = p$

Expected payoff of acquiescing after U is $2p$ and of fighting is $1 - 2p \Rightarrow A$ after U if $p \geq \frac{1}{4}$

- Incumbent chooses A after U. Strong Challenger gets 5 from U. Regardless of Incumbent's strategy, R is not a good deviation. There are PBE in which both types choose U, Incumbent chooses A after U
- Incumbent chooses F after U. Strong challenger gets 3 from U. If it deviates to R, payoff is 2 if F and 4 if A . To sustain PBE, Incumbent must play F after R, which requires Challenger to be sufficiently likely to be Weak: $\mu(R) \leq \frac{1}{2}$. Are these beliefs consistent? Yes because off equilibrium path

Separating and Pooling Equilibria

This example illustrates the two kinds of pure strategy equilibria that can exist in signaling games:

Definition

A **(fully) separating equilibrium** of a signaling game is one in which each type of sender chooses different actions

[Is this always possible?]

- Upon observing the sender's action, the receiver knows the sender's type

Definition

A **pooling equilibrium** of a signaling game is one in which all types of the sender choose the same strategy

Exercise

Show that if one modifies one payoff to one player in one terminal node of the crisis bargaining game, the separating equilibrium no longer exists

What is a signaling game

An extensive form of incomplete information

- Two players: $N - \{S, R\}$
- A payoff payoff-relevant state $t \in T$ drawn from a distribution $f(\cdot)$
- only S observes t (so we refer to t as S's type), so $\tau_S(t) = t$, $\tau_R(t) = \emptyset \; \forall t \in T$
- After observing t, S sends a signal $s \in M$ to R
- R, after observing s, takes action $a \in A$
- A terminal history is a triplet (state, signal, R's action), so $u_i : T \times M \times A \to \mathbb{R}$

(Perfect Bayesian) Equilibrium

Assessment (α, σ, μ) form an Equilibrium iff

- S's strategy maximizes expected payoff given $\alpha(\cdot, \cdot)$: for all $t \in T$, $\sigma(s, t) > 0$ implies

$$\sum_{a \in A} u_S(t, s, a)\alpha(a, s) = \max_{s' \in M} \sum_{a \in A} u_S(t, s', a)\alpha(a, s')$$

- R's strategy maximizes her expected payoff given $\mu(\cdot, s)$: $\forall s \in M$, $\alpha(a, s) > 0$ implies

$$\sum_{t \in T} u_R(t, s, a)\mu(t, s) = \max_{a' \in A} \sum_{t \in T} u_R(t, s, a')\mu(t, s)$$

- $\mu(\cdot, s)$ is consistent with σ: if $\sigma(s, t) \neq 0$ for some $t \in T$,

$$\mu(t, s) = \frac{\sigma(s, t) f(t)}{\sum_{t' \in T} \sigma(s, t') f(t')}$$

Separating vs Pooling

Consider an equilibrium $(\alpha^*, \sigma^*, \mu^*)$

- In a separating equilibrium (α^*, σ^*), the message space can be partitioned into a collection of $\{M_t\}_{t \in T \cup \emptyset}$ such that for each t, $\sum_{s \in M_t} \sigma^*(s,t) = 1$
- In a pooling equilibrium, there exists a set $M_p \subset M$ such that $\forall\ s \in M_p$, $\sigma^*(s,t) = \sigma^*(s,t')\ \forall (t,t') \in T^2$

The existence of a separating equilibrium

Suppose that T is an ordered set, M, A are real intervals

Definition

$u_S(\cdot)$ satisfies the **single crossing property** if $u_S(t, s', a') \geq u_S(t, s'', a'')$ for $s' > s''$ implies

$$u_S(t', s', a') > u_S(t', s'', a''), \quad \forall t' > t$$

Indifference curves for sender in $M \times A$ are well defined

Single crossing says that indifference curves of different types of senders cross (at most) once

NB: can extend idea to partially ordered domain for u_S, as well as mixed strategy for a

The existence of a separating equilibrium

Definition

In a **standard** signaling game,[1]

- $T = \{0, \cdots, K\}$, $A = [\underline{a}, \overline{a}]$ and $M = [\underline{s}, \overline{s}]$ are real intervals
- utility functions are continuous in actions and signals
- $u_S(\cdot)$ is strictly increasing in action, strictly decreasing in signal, and the single crossing property holds
- u_R is independent of s, R's best response $BR(t)$ is uniquely defined and strictly increasing in t
- No "bunching" $u_S(K, \overline{s}, BR(K)) < u_S(K, \underline{s}, BR(0))$

Proposition

Proposition 1. *The standard signaling game admits a separating equilibrium*

1. The last two conditions are technical, the first two can be generalized

Exercise

Verify that the following is a standard signaling game:

- $K = 5$, $A = [0, 1], M = [0, 10]$
- $u_R(t, s, a) = -\left(a - \frac{t}{K}\right)^2$, $u_S(s, t, a) = a - \frac{s}{1+t}$

Refinements

Equilibrium refinements (stronger than Nash) also make assumptions on beliefs following signals that are never played by any type

- That is, $s \in M : \sum_{t \in T} \sigma(s,t) f(t) = 0$

For example, sequential equilibrium essentially requires that condition (1) holds even for these values of s

- The idea is to rule out equilibria in which some s' is not sent because R's response is a dominated action
- More on this later

Education as a Signal

The canonical example of a game with costly signals is the model of education acquisition due to Spence

Players: Two firms (1 and 2) and one worker

Worker's ability is $\theta \in \{\theta_L, \theta_H\}$, where $0 = \theta_L < \theta_H$. Ability is worker's private information

Timeline

1. Nature chooses worker's type (θ_H with probability π, θ_L otherwise)
2. Worker chooses an amount of education ($e > 0$)
3. Firms observe education and offer wages w_1, w_2
4. Worker accepts one of the two offers

Payoffs and Strategies

Worker likes wages, suffers cost of education:

$$u_i(\theta_i, e, w) = w - c(e, \theta_i)$$

If firm j hires a worker with ability θ_i, at wage w, its payoff is

$$u_j(\theta_, w) = f(\theta_i) - w$$

For simplicity, let $f(\theta_i) = \theta_i$. Notice that education does not improve productivity!

Assume $c_e > 0$, $c_{ee} > 0$, and normalize $c(0, \theta) = 0$

Assume single crossing: $c_{e\theta} < 0$, i.e., **increasing educational attainment is “cheaper” for high-ability types**

Strategies and equilibria

A (pure strategy) assessment: $\{e_L, e_H, w_1(e), w_2(e), \mu_H^1(e), \mu_H^2(e)\}$

For $e \in \{e_L, e_H\}$, let $\mu_H(e) \in \{0, \pi, 1\}$ the consistent belief

Claim: If there is a PBE with $w_1(e) \neq w_2(e)$, there is an outcome-equivalent PBE with $w_1(e) = w_2(e) = \mu_H(e)\theta_H$

Proof.

Proof. (1.) Firms are allowed to have different beliefs and strategies off-path (i.e., for $e \notin \{e_L, e_H\}$), it does not affect outcomes

(2.) Suppose wlog that for some $e \in \{e_L, e_H\}$, $\max\{w_1(e), w_2(e)\} > \mu_H(e)\theta_H$. Then at least one firm is making negative profits and has a strict deviation to $\mu_H(e)\theta_H$

(3.) If $\max\{w_1(e), w_2(e)\} < \mu_H(e)\theta_H$, at list one firm can increase its profits by deviating to $\max\{w_1(e), w_2(e)\} + \epsilon$ □

Hence, focus on assessments with $w_1(e) = w_2(e) = w(e)$ and $\mu_{1H}(e) = \mu_{2H}(e) = \mu_H(e)$

Separating Equilibrium

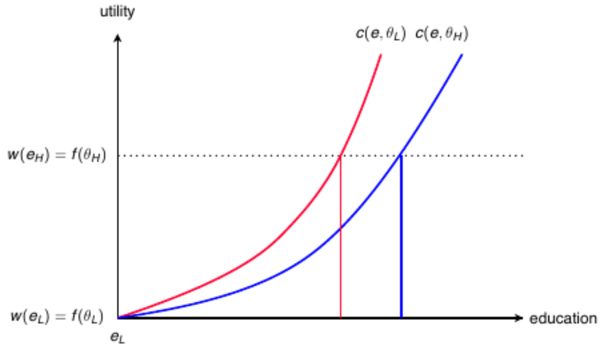
Suppose we are in a separating equilibrium. Then

- Firms learn type from education: $\mu_H(e_H) = 1$, $\mu_H(e_L) = 0$
- Wage offers equal productivity: $w(e_H) = \theta_H$, $w(e_L) = \theta_L = 0$
- $e_L = 0$, otherwise profitable deviation for θ_L:
$$u_S(\theta_L, e_L, 0) = -c(e_L, \theta_L) < 0 \leq w(0) = u_S(\theta_L, 0, w(0))$$
- $e_H \in [\underline{e}, \overline{e}]$, with
$$\begin{cases} \theta_H = c(\overline{e}, \theta_H) & (e_H, \theta_H) \succsim_{\theta_H} (0,0) \\ \theta_H = c(\underline{e}, \theta_L) & (e_H, \theta_H) \precsim_{\theta_L} (0,0) \end{cases}$$
If $e < \underline{e}$, θ_L prefers e_H, if $e > \overline{e}$, θ_H prefers $e_L = 0$
- What about $e' \notin \{0, e_H\}$? Assume $\mu_H(e') = 0$ (PBE allows it)
$\Rightarrow w(e') = 0$ and e' is strictly dominated for both types

Separating Equilibrium (cont'd)

A continuum of separating equilibria with $e_L = 0$, $e_H \in [\underline{e}, \overline{e}]$,

$$(w(e), \mu_H(e)) = \begin{cases} (\theta_H, 1) & e = e_H \\ (0, 0) & \text{otherwise} \end{cases}$$



Pooling

In a pooling equilibrium, all types choose the same level of education e^*

Employers do not learn type, so $\mu = \pi$

Wage will be $w^* = \pi\theta_H$

Assume that, after seeing $e < e^*$, firms believe worker is low type, and after seeing $e > e^*$, firms stay at prior

$$\mu(e) = \begin{cases} 0 & e < e^* \\ \pi & e > e^* \end{cases}$$

Pooling (cont'd)

Clearly, no one will deviate up

We also need no deviation down:

$$\pi\theta_H - c(e^*, \theta_L) > 0$$

and

$$\pi\theta_H - c(e^*, \theta_H) > 0$$

The second condition doesn't bind, since $c(e^*, \theta_L) > c(e^*, \theta_H)$.

Hence, pooling requires that

$$c(e^*, \theta_L) < \pi\theta_H$$

Multiplicity

Multiplicity problem is typically profound, with a continuum of distinct equilibrium outcomes (when M is an interval)

Without refinement, equilibrium theory provides few clear predictions beyond

- the lowest type of sender receives at least her complete information payoff
- the equilibrium signaling function $s^*(t)$ is weakly increasing in the sender's type

Exercise Explain why the two statements above must be true

Can we make sharper predictions?

A refinement: Sequential Equilibrium

Back to general signaling game notation

Let B^0 denote the set of fully mixed strategy profiles of a signaling game

$$B^0 = \left\{ (\alpha, \sigma) : \alpha(a, s) > 0 \ \forall (a, s) \text{ and } \sigma(s, t) > 0 \ \forall (s, t) \right\}$$

A belief system μ satisfies S-consistency with (α, σ) if there exists a fully mixed sequence $\{\alpha^n, \sigma^n\}$ in B^0 such that

$$(\alpha, \sigma, \mu) = \lim_{n \to \infty} (\alpha^n, \sigma^n, \mu^n)$$

where μ^n is the unique belief system consistent with (α^n, σ^n).

Definition

A sequential equilibrium is a Perfect Bayesian Equilibrium in which the belief system also satisfies S-Consistency.

Illustration

Suppose nature chooses $\theta \in \{h, l\}$, with $\Pr(\theta = h) = \pi$

Then Pl. 1 chooses $b \in \{U, D\}$ without observing θ

Then Pl. 2 observes b and forms belief $\mu_h(b) = \Pr(\theta = h)$

Now suppose that Pl. 1's strategy is such that $\sigma_1(U) = 1$

Consistency places no restrictions on $\mu_h(b)$

S-consistency does. Any $\{\sigma_1^n(U)\}$ with $\sigma_1^n(U) \xrightarrow[n\to\infty]{} 1$ yields

$$\mu_h^n(D) = \frac{\Pr(b = D|\theta = h))\Pr(\theta = h)}{\Pr(b = D|\theta = h))\Pr(\theta = h) + \Pr(b = D|\theta = l))\Pr(\theta = l)}$$
$$= \frac{(1 - \sigma_1^n(U))\gamma}{1 - \sigma_1^n(U)} = \gamma$$

A refinement: The Intuitive Criterion

Given an equilibrium $\psi^* = (\alpha^*, \sigma^*, \mu^*)$, let $U(\psi^*, t)$ be the equilibrium expected payoff of type t and let

$$T^{NED}(s; \psi^*) = \{t \in T : \max_{a \in BR(s)} U(a, s, \mu, t) \geq U(\psi^*, t)\}$$

where $BR(s)$ satisfies condition (2) for *some* belief μ

$T^{NED}(s; \psi^*)$: types for which s is Not Equilibrium Dominated

Definition

ψ^* satisfies the Intuitive Criterion if for any unsent message s, $\mu(\cdot, s)$ places probability one on $t \in T^{NED}(s; psi^*)$

Any $t \notin T^{NED}(s; psi^*)$ has no reason whatsoever to choose s.

In Spence model, IC requires $\mu(e) = 1$ for all $e > \underline{e}$

A refinement

Cho and Kreps introduce a refinement called D1

Given an equilibrium ($\alpha^*, \sigma^*, \mu^*$), let $U^*(t)$ be the equilibrium expected payoff of a type t sender, and let $D(s;t) = \{a : u_S(t, s, a) \geq U^*(t)\}$ be the set of pure-strategy responses to s that lead to payoffs at least as great as the equilibrium payoff for player t. Given a collection of sets, $\{X(t)\}_{t \in T}$, $X(t^*)$ is maximal if it not a proper subset of any $X(t)$

Definition

Behavior strategies (α^*, σ^*) together with beliefs μ^* satisfy D1 if for any unsent message s, $\mu(\cdot, s)$ places positive probability on those t for which $D(s, t)$ is maximal

If $D(s, t)$ is not maximal, another type t' is *more likely to deviate*: there exist out-of-equilibrium responses that are attractive to t' but not t. D1 requires that the receiver upon observing s places no weight on type t

Not going to prove this

Proposition

Proposition 2. *The standard signaling game has a unique separating equilibrium outcome that satisfies Condition D1*